Primary 3 · Science

Science · Set 5 — Answer Key

80 marks · 1 h 15 min · Challenge

For parents. Give this to the pupil only after the paper is finished. Written answers: accept any answer with the same meaning, in a complete sentence with correct tense.

Section A: Multiple-Choice Questions (40 marks)

No.	Answer	Marks
1	(1) The type of soil	2
2	(1) The amount of water poured and the size of the pot	2
3	(1) Soil A	2
4	(4) The type of soil	2
5	(3) New seeds	2
6	(1) Seed X	2
7	(4) A pot of basil growing on a windowsill	2
8	(3) Mammals	2
9	(3) Guinea pig	2
10	(4) The towel was kept damp and in the dark	2
11	(3) Mosquito and beetle	2
12	(3) Cockroach	2
13	(4) Egg → Tadpole → Tadpole with legs → Adult frog	2
14	(2) Larva	2
15	(2) Waterproof	2
16	(2) Double-walled insulated steel	2
17	(4) A closed empty plastic bottle	2
18	(1) A growing crystal	2
19	(2) Both hatch from eggs	2
20	(2) Is a living fungus that feeds and produces gas as it grows	2

Section B: Structured Questions (16 marks)

No.	Answer	Marks
21(a)	False	1
21(b)	True	1
22	(1) The force used to push the car	2
23	Does not get hot easily (it is a poor conductor of heat / a good insulator), protecting the hand from the hot pan.	2
24	Flexible — it can bend easily, so the ice cubes pop out without cracking the tray.	2
25	(2) The orange's skin slowly broke down and stayed moist	2
26(a)	False	1
26(b)	True	1
27(a)	True	1
27(b)	False	1
28	It cannot feed, breathe, respond to its surroundings, or reproduce, the way a living thing can (accept any one).	2

Section C: Open-Ended Questions (24 marks)

No.	Answer	Marks
29	Only the type of material was changed. The size of each material, the amount of water, and the time each one was dipped in the water were kept the same. This makes it a fair test because any difference in how much water is absorbed is caused only by the type of material, not by any other factor. — Marking guide: 1 mark for correctly stating what was changed (the type of material), 1 mark for correctly stating what was kept the same (size of material, amount of water, dipping time), 1 mark for explaining why this makes the test fair.	3

30	Only the material the ice cube was wrapped in was changed. The size of the ice cube and the room temperature were kept the same for every ice cube. This makes it a fair test because any difference in how long the ice takes to melt is caused only by the wrapping material, not by any other factor. — Marking guide: 1 mark for correctly stating what was changed (the wrapping material), 1 mark for correctly stating what was kept the same (size of ice cube, room temperature), 1 mark for explaining why this makes the test fair.	3
31	She could gently touch the snail's body to see if it pulls back into its shell or responds, and she could watch closely over a few minutes to see if it slowly moves or if its breathing hole opens and closes. — Marking guide: 1 mark for a sensible test of response to touch/stimulus, 1 mark for a second sensible test (e.g. checking for slow movement or breathing), 1 mark for a clear, complete explanation. Give 0 marks for actions that are CARE rather than a TEST (e.g. giving it food, or moving it to a damp spot).	3
32	Mould is a fungus, so it cannot make its own food using sunlight. Instead, it feeds on the damp sponge as a source of food. Keeping food covered and dry removes the moisture and food source that mould needs to grow, so mould is much less likely to grow on it. — Marking guide: 1 mark for stating that mould (a fungus) feeds on the damp material rather than making its own food, 1 mark for linking mould's growth to needing moisture and a food source, 1 mark for the missing link — explaining why keeping food covered and dry stops mould from growing.	3
33	A dragonfly has no pupa stage — its nymph already looks somewhat similar to the adult and gradually grows bigger and develops wings after each moult. A beetle, however, goes through a pupa stage, during which its body changes completely from a larva into an adult, which the dragonfly never goes through. — Marking guide: 1 mark for describing the dragonfly nymph as gradually developing (no pupa, no big sudden body change), 1 mark for describing the beetle's pupa stage as a stage of major body change, 1 mark for a clear comparison between the two.	3
34	A double-walled steel flask is a good choice because the trapped air between its two walls does not let heat escape easily, keeping the drink hot for longer. It is also strong and does not crack or break easily, unlike materials such as thin plastic or glass, so it can be carried around safely. — Marking guide: 1 mark for explaining that the trapped air/double wall keeps heat in (slows heat loss), 1 mark for a second correct property (e.g. strong, does not break easily), 1 mark for a clear overall explanation linking the properties to why it suits a flask.	3

35	Although both mould and algae can appear and grow without being deliberately planted, the key difference is that algae, like other plants, can make their own food using sunlight, but mould, which is a fungus, cannot make its own food — it gets its food by feeding on the bread as it decays instead. — Marking guide: 2 marks for stating the key difference clearly for BOTH sides — fungi like mould cannot make their own food, but plants like algae can make their own food using sunlight (1 mark for each side). 1 mark for noting that appearing without being planted does not make them the same kind of living thing.	3
36	A seed needs a suitable (warm enough) temperature to germinate, as well as water and air. Pot X was kept at room temperature, which was suitable, so the seed was able to germinate. Pot Y was too cold inside the refrigerator, so even though it had water and air, the seed could not germinate. — Marking guide: 1 mark for stating that a seed needs a suitable temperature to germinate (besides water and air), 1 mark for linking Pot X's germination to its suitable room temperature, 1 mark for the missing link — explaining that Pot Y's seed did not germinate because the refrigerator was too cold, even though it had water and air.	3